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*Mol Genet Metab*. Author manuscript; available in PMC 2020 September 01.

Published in final edited form as:

*Mol Genet Metab*. 2019 ; 128(1-2): 144–150. doi:10.1016/j.ymgme.2019.05.015.

# Urinary 2,8-Dihydroxyadenine Excretion in Patients with APRT Deficiency, Carriers and Healthy Control Subjects

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## Abstract

**Background:** Adenine phosphoribosyltransferase (APRT) deficiency is a rare autosomal recessive disorder of adenine metabolism that results in excessive urinary excretion of the poorly soluble 2,8-dihydroxyadenine (DHA), leading to kidney stones and chronic kidney disease. The purpose of this study was to assess urinary DHA excretion in patients with APRT deficiency, heterozygotes and healthy controls, using a recently developed ultra-performance liquid chromatography tandem mass spectrometry (UPLC-MS/MS) assay.

**Methods:** Patients enrolled in the APRT Deficiency Registry and Biobank of the Rare Kidney Stone Consortium (RKSC; http://www.rarekidneystones.org/) who had provided 24-hour and first-

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**Details of the contributions of individual authors:** Conception and study design: VE, RP, HLR, OSI; laboratory work and DHA measurements: UAT, MTh, GSO; statistical analysis: HLR, OSI; critical review of results: HLR, IMA, OSI, RP, VE, UAT, MTh; supervision and mentorship: VE, RP, OSI. HLR and VE wrote the first manuscript draft. Each author contributed important intellectual content during manuscript drafting and accepts accountability for the overall work by ensuring that questions pertaining to the accuracy or integrity of any portion of the work are appropriately investigated and resolved.

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**Competing Interest Statement: HLR, IMA, OSI, RP, SGO, VE and UAT** declare no financial or other competing interests. **MT** owns stock in ArcticMass.

**Data Availability and Material Statement** Data used for this project are kept in the APRT Deficiency Registry of the Rare Kidney Stone Consortium. Biological samples used were stored in the Rare Kidney Stone Consortium Biobank, housed at Landspitali–The National University Hospital of Iceland in Reykjavik.

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morning void urine samples for DHA measurements, were eligible for the study. Heterozygotes and healthy individuals served as controls. Wilcoxon-Mann-Whitney test was used to compare 24-hr urinary DHA excretion between groups. Associations were examined using Spearman's correlation coefficient.

**Results:** The median 24-hr urinary DHA excretion was 138 (64–292) mg/24 hr and the DHA/Cr ratio in the first-morning void samples was 13 (4–37) mg/mmol in APRT deficiency patients who were not receiving xanthine oxidoreductase inhibitor therapy. The 24-hr DHA excretion was highly correlated with the DHA/Cr ratio in first-morning void urine samples ($r_s$ = 0.84, p <0.001). DHA was detected in all urine samples from untreated patients but not in any specimens from heterozygotes and healthy controls (p=.000).

**Conclusions:** High DHA excretion was observed in patients with APRT deficiency, while urine DHA was undetectable in heterozygotes and healthy controls. Our results suggest that the UPLC-MS/MS assay can be used for diagnosis of APRT deficiency.

### Keywords

Chronic kidney disease; Crystalluria; Kidney stones; Timed urine collections; First-morning void urine samples; UPLC-MS/MS assay; Diagnosis of APRT deficiency

## Introduction

Adenine phosphoribosyltransferase (APRT) deficiency (OMIM 102600) is a rare autosomal recessive disorder of purine metabolism that results in excessive renal excretion of the poorly soluble 2,8-dihydroxyadenine (DHA), leading to kidney stones, acute kidney injury (AKI) and DHA crystal-induced crystal nephropathy and chronic kidney disease (CKD) (1–3). Radiolucent kidney stones are the most common clinical manifestation of APRT deficiency (1, 3), but 15-20% of adult cases present with end-stage kidney disease (ESRD) secondary to crystal nephropathy (2). The xanthine oxidoreductase (XOR; xanthine dehydrogenase/oxidase) inhibitors allopurinol and febuxostat effectively reduce urinary DHA excretion (4) and have been found to ameliorate disease manifestations, including kidney stone formation and progressive CKD (1–3). While the typical round and brown urine DHA crystals are highly suggestive of APRT deficiency, abolished APRT enzyme activity in red cell lysates and/or identification of biallelic mutations in the APRT gene have been required for definite diagnosis of the disorder (5).

Therapeutic drug monitoring is traditionally performed by urine microscopy where the absence or at least reduction of DHA crystals is thought to indicate adequate XOR inhibitor treatment (5). Several methods for quantifying DHA have been described, including techniques based on HPLC coupled to tandem mass spectrometry and a multichannel ultraviolet detector, as well as capillary electrophoresis (6–10). Our group recently developed a ultra-performance liquid chromatography – electrospray tandem mass spectrometry (UPLC-MS/MS) assay with isotopically labeled internal standard for absolute urinary quantification of DHA (4, 11).

The aims of the present work were to use a recently developed UPLC-MS/MS urinary assay to measure urinary DHA excretion in APRT deficiency patients, heterozygotes and healthy

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control subjects, to correlate the 24-hr urinary DHA excretion with the DHA-to-creatinine (DHA/Cr) ratio in single-void urine specimens and to test the sensitivity and specificity of the UPLC-MS/MS urinary DHA assay for the diagnosis of APRT deficiency.

## Methods

### Ethical approval

The study was approved by the Icelandic National Bioethics Committee (NBC 09-072 and 13-115-S1) and the Icelandic Data Protection Authority. All study participants gave written informed consent for their participation. The clinical and research activities reported are consistent with the ethical principles of the Declaration of Helsinki.

### Study subjects and clinical data

Patients in the APRT Deficiency Registry and Biobank of the Rare Kidney Stone Consortium (RKSC; http://www.rarekidneystones.org/) who had donated urine samples for DHA measurement before April 30, 2018 were eligible for participation in the study. Individuals heterozygous for pathogenic *APRT* mutations and healthy subjects who were not taking medications affecting the metabolism or excretion of purines served as controls. Registry data included clinical manifestations, medication use (XOR inhibitors, allopurinol or febuxostat), height, weight, serum creatinine (SCr), urine creatinine (UCr) and results of APRT activity measurements and genetic testing. The creatinine measurements were traceable to a reference method based on isotope dilution mass spectrometry (IDMS). The CKD-EPI equation (12) was used to estimate glomerular filtration rate (eGFR) in adults and the CKiD Schwartz equation in patients < 18 years of age (13).

### Measurement of urine 2,8-dihydroxyadenine

Biobanked 24-hr urine specimens and single-void urine samples were used for the study. All participants were on their habitual diet when the urine samples were collected. Twenty-four hour urinary DHA excretion was measured in patients with APRT deficiency, both on and off XOR inhibitor treatment, and in control subjects. Correlation between the DHA/Cr ratio in first-morning void urine specimens and 24-hr urinary DHA excretion, both obtained in the same 24-hr period, was assessed in all available sample pairs, regardless of XOR inhibitor treatment status.

Urine DHA was measured using the UPLC-MS/MS assay developed by our group and expressed as mg/24 hr in timed samples and DHA/Cr ratio in mg/mmol in single-void specimens as previously described(11). The lower limit of DHA detection and quantification were 20 and 100 ng/mL, respectively. Urinary DHA excretion values based on timed urine samples were adjusted for collection time (median collection time 1375 (870-1670) min) and standardized to 1440 minutes (24 hr). DHA crystalluria detected by urine microscopy was semiquantitatively graded in our laboratory as 0, (+),1+, 2+, 3+ and 4+ by an experienced medical laboratory scientist before the urine specimens were frozen for storage.

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### Statistical analyses

Data are presented as numbers or median (range). Wilcoxon-Mann-Whitney test was used to compare 24-hr urinary DHA excretion between groups. When participants had more than one 24-hr urine sample available, the mean urine DHA excretion for each individual was used in the analysis. Spearman's ($r_s$) correlation coefficient was used to assess the correlation between 24-hr urine DHA excretion and the first-morning void DHA/Cr ratio and DHA crystalluria, as well as weight, body surface area (BSA) and eGFR. Outliers were identified with visual examination of the scatter-plot. The sensitivity and specificity of the UPLC-MS/MS urinary DHA assay for diagnosis of APRT deficiency was examined by comparing the urinary DHA concentration in samples from patients, carriers and healthy controls.

## Results

### Characteristics of participants

Thirty-three of the 60 patients in the APRT Deficiency Registry, 22 of whom were women, had provided urine samples for DHA measurement. Most patients had multiple samples available; there were a total of 91 24-hr urine collections, 182 first-morning and 24 random-void urine specimens (Table 1). A single 24-hr urine and a first-morning void urine sample pair, obtained in the same 24-hr period, had been obtained from 4 heterozygotes and 10 healthy control subjects for DHA measurement. The number and type of urine samples available for each analysis is noted below.

Twenty-one of the 33 patients had a past history of kidney stones, 11 had experienced AKI episodes and 8 patients had developed CKD stage 5, one of whom was on dialysis and 6 had undergone kidney transplantation. Six patients had an asymptomatic course. At the time of last urine sampling, one patient had eGFR of 14 mL/min/1.73m$^2$ when a first-morning void urine sample was obtained, while another had eGFR 40 mL/min/1.73m$^2$ at the time of a 24-hr urine collection. All other patients had eGFR >60 mL/min/1.73m$^2$.

### Urinary 2,8-dihydroxyadenine excretion

Thirty-one timed urine samples from 14 untreated patients (10 women) with APRT deficiency were available for the determination of DHA excretion (Table 1). The median (range) 24-hr urinary DHA excretion was 138 (64–292) mg, while DHA was not detected in any 24-hr urine specimens from the 4 heterozygotes and 10 healthy controls (Table 2). The urinary DHA excretion was significantly greater in men, 233 (95-289) mg/24 hr compared with 129 (64-291) mg/24 hr in women (p = .03; Figure 1A), which did change when corrected for weight or BSA. No association was observed between DHA excretion and weight ($r_s$= 0.36, p = .203), BSA ($r_s$ = 0.36, p= .208), eGFR ($r_s$ = 0.35; p = .227; Figure 1B–D) or age ($r_s$= −0.332, p = 0.068). A marked intra-individual variability in urinary DHA excretion was observed among 5 untreated patients who had multiple urine collections available at different time points (Figure 2). The DHA/Cr ratio was determined in 44 first-morning void urine samples available from 19 untreated patients and 24 random-void urine samples from 4 untreated patients, yielding a median of 12.7 (3.8–37.2) mg/mmol and 15.5 (10.4-19.3) mg/mmol, respectively. A total of 121 first-morning void urine samples were

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available from 33 patients while they were receiving XOR-inhibitor treatment. No DHA was detected in 44 of these samples (from 25 patients), whereas the DHA/Cr ratio was 4.5 (0.4-24.8) mg/mmol in 77 samples (from 21 patients).

### Correlation of 2,8-dihydroxyadenine in timed and single-void urine specimens

Thirty-six pairs of 24-hr and first-morning void urine samples collected by 11 patients in the same 24-hr period, were available for analysis of the correlation between the DHA/Cr ratio in first-morning void urine specimens and 24-hr DHA excretion (Table 1). The correlation was highly significant, both before ($r_s$ = 0.78, p <0.001) and after removing 3 outliers ($r_s$ = 0.84, p <0.001; Figure 3). Seventeen of these 36 sample pairs were obtained during treatment with an XOR inhibitor and 19 off pharmacotherapy.

### Correlation between 24-hr 2,8-dihydroxyadenine excretion and crystalluria

A highly significant correlation was observed between the 24-hr urinary DHA excretion and DHA crystalluria on urine microscopy ($r_s$=0.810, $p < 0.001$; Figure 4) in 91 urine sample pairs (60 obtained during XOR inhibitor treatment and 31 off pharmacotherapy) available for that analysis (Table 1). Interestingly, urinary DHA excretion in the range of 19–88 mg/24 hr was noted in 16 of the 40 urine samples that were deemed negative for DHA crystals. All 16 samples were from patients treated with allopurinol in a daily dose of 300-400 mg, while the remaining 24 timed urine samples were collected during treatment with febuxostat 80 mg/day and had undetectable levels of DHA.

To test the accuracy of the UPLC-MS/MS urinary DHA assay for the diagnosis of APRT deficiency, we included a total of 99 urine samples (24 hr urine collections, n=31; first-morning void urine, n=44; random void urine, n=24) from 21 untreated patients and 28 specimens (24 hr urine collection, n=14; first-morning void urine, n=14) from 14 unaffected controls. Urine DHA was detected in all patient specimens, but not in any of the samples from the 4 heterozygotes and the 10 healthy control subjects. Thus, the detection of urine DHA using the novel UPLC-MS/MS assay confers 100% sensitivity and specificity for the the diagnosis of APRT deficiency in patients who are not receiving XOR inhibitor treatment.

## Discussion

In the current study, all patients with APRT deficiency had high, albeit variable, urinary DHA excretion that was significantly greater in men than women. A strong correlation was observed between DHA excretion in timed urine samples and the DHA/Cr ratio in first-morning void urine specimens. Importantly, urinary DHA was not detected in samples from heterozygotes, healthy individuals and many treated patients.

Our findings demonstrate a wide range of urinary DHA excretion in untreated patients with APRT deficiency. While intra- and inter-day variability in the accuracy of quantification associated with our UPLC-MS/MS DHA assay is within ±15% (11), it does not explain the large variability in urinary DHA excretion seen in our patient cohort. A more likely explanation is variability in dietary or systemic adenine load and/or the rate at which it is converted to DHA by XOR. Moreover, multiple types of human intestinal bacterial taxa use APRT to metabolize adenine to adenosine monophosphate through adenine and adenosine

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salvage pathways (14, 15). Therefore, modulation of adenine metabolism by the gut microbiome, affecting the amount of adenine available for intestinal absorption, may have contributed to the variable urinary DHA excretion observed between study subjects. High-purine diet may also increase systemic adenine load by providing ample amounts of the substance for intestinal absorption. One single-patient report showed a decrease in urinary DHA excretion on a purine-restricted diet when compared with a high-purine diet, suggesting a significant contribution of dietary adenine intake to the systemic adenine supply (16). Rodents fed adenine-enriched diet are well known to develop adenine or DHA nephropathy within weeks, a form of kidney damage closely mimicking human DHA nephropathy (17). Finally, significant inter-individual differences have recently been reported in plasma XOR activity in humans (18), which might be a plausible mechanism for the variability in renal DHA excretion observed in our patient cohort.

No obvious explanation exists for the significantly higher urinary DHA excretion observed in men compared with women, a finding that has not been reported previously. The larger muscle mass in men may have contributed to some of the differences observed, although we did not find any significant variability in DHA excretion when correlated to BSA. The generally larger food portions consumed by men may also have contributed to the sex differences, although we have no data to support this notion. Interestingly, there are reports showing differences in the intestinal microbiome between the sexes(15), a fact that may have implications for the difference in DHA excretion observed between men and women. The small number of men in our study may have amplified any errors associated with urine sampling, such as inaccuracy in the documentation of collection time, which may have falsely increased the DHA excretion. Additionally, also due to the small sample size, the differences in urinary DHA excretion between the sexes may simply have occurred by chance.

The intra-individual variability noted in the current study in DHA excretion could also be affected by sampling or measurement errors, while temporal changes in systemic adenine load, and less likely XOR activity, cannot be excluded. Although we did not find any correlation between urinary DHA excretion and eGFR, it is important to note that only one untreated patient had severe renal dysfunction at the time of urine sampling.

Urinary DHA excretion rates have previously only been published in a few cases of APRT deficiency and the reported values in both random void (7) and timed (9, 16) urine specimens, have been at the very low end or markedly below the range observed in our patients. While this may be explained by dietary factors or other influences mentioned earlier, methodological differences in the measurement methods used may also play a role. Our recently developed UPLC-MS/MS urinary assay uses an isotope-labeled internal standard for absolute DHA quantification which has been lacking in previously reported techniques. The isotope-labeled internal standard, an important feature of our method, corrects for errors which might be present during sample preparation and analysis.

The close correlation between 24-hr urinary DHA excretion and the DHA/Cr ratio in first-morning void urine samples in the present study is an important finding. Based on these results, the determination of 24-hr urinary DHA excretion can be replaced with DHA/Cr

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ratio in first-morning void urine samples for monitoring the effect of pharmacotherapy and treatment adherence, both in the clinic and clinical research studies. Assessment of the excretion of various urinary biomarkers has been challenging through the years, due in part to the tedious nature of 24-hr urine sampling which has rendered sample collections subject to poor compliance and error. Hence, the measurement of solute/Cr ratio in random urine samples has been increasingly used for quantification of renal excretion. The best example is first-morning void protein/Cr ratio which has been shown to correlate well with 24-hr urinary albumin excretion, the correlation coefficient being over 0.8 in most reports (19–21), as in the present study.

A strong correlation was also observed between 24-hr urinary DHA excretion and DHA crystalluria, currently widely used for therapeutic monitoring. For example, absence of crystalluria mostly correlated with unquantifiable amounts of DHA. However, it is noteworthy that significant urinary DHA excretion, up to 88 mg/24 hr, was observed in several timed urine collections despite the absence of microscopic crystalluria, whereas DHA levels in this range were generally found in samples with 1-3+ DHA crystalluria. This finding suggests that DHA crystalluria is not reliable enough for assessment of the effectiveness of XOR inhibitor therapy. Currently, it is not known if persistent DHA excretion in low levels enhances the risk of kidney disease progression. Thus, while assessment of crystalluria for guiding pharmacotherapy is useful, it should be used with a degree of caution in the absence of a more precise method.

The gold standard methods for the diagnosis of APRT deficiency, enzyme activity measurements and genetic testing, are cumbersome and not widely available. Hence, a more rapid method for screening and diagnosis of this rare and frequently underdiagnosed disease is of great value. Since urine DHA was undetectable in heterozygotes and healthy control subjects and abundant in untreated patients, our UPLC-MS/MS method for absolute quantification of urinary DHA excretion appears to be highly accurate in identifying patients with APRT deficiency. Indeed, our findings suggest that detection of any urine DHA using our assay can be considered diagnostic of APRT deficiency in individuals who have not recently received large parenteral adenine load as occurs with massive blood transfusions (22).

Strengths of our study include the relatively large number of urine samples available from this rare disease population, including timed urine collections. Another important advantage is the use of our highly sensitive and accurate UPLC-MS/MS assay that includes an isotope-labeled internal standard, allowing for absolute quantification of urinary DHA concentration. Nevertheless, the study is limited by a small number of participants although this would be expected for a rare disease. The small number of patients with advanced CKD limited our ability to assess the effect of reduced kidney function on DHA excretion. Moreover, information on dietary intake was neither available for patients nor controls.

In conclusion, high urinary DHA excretion was observed in all patients with APRT deficiency and the excretion was greater in men than women. The strong correlation between 24-hr urinary DHA excretion and DHA/Cr ratio in first-morning void samples suggests that timed collections may be replaced by first-morning void samples in monitoring of the

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effectiveness of XOR inhibitor therapy and treatment adherence. DHA was not detected in urine samples from heterozygotes and healthy individuals suggesting that our robust and reliable UPLC-MS/MS urinary DHA assay can be added to the list of diagnostic tests for APRT deficiency. Future studies should focus on factors that might affect production and urinary excretion of DHA in patients with APRT deficiency, as well as the impact of different levels of DHA excretion on the progression of CKD.

## Acknowledgements

Part of this work was presented in an abstract form at the American Society of Nephrology Kidney Week, October 23-29, 2018, in San Diego, CA. The authors wish to sincerely thank the following physicians for their invaluable assistance in clinical data and biosample collection: Dawn Milliner (Mayo Clinic, Rochester, MN, USA), John Lieske (Mayo Clinic, Rochester, MN, USA), David Goldfarb (New York University, New York, NY, USA) and James S. Shawcross (North Cumbria University Hospitals, Carlisle, UK).

Funding Support

This study was supported by the Rare Kidney Stone Consortium (2U54DK083908), a part of the National Center for Advancing Translational Sciences (NCATS) Rare Diseases Clinical Research Network (RDCRN). RDCRN is an initiative of the Office of Rare Diseases Research (ORDR). The Rare Kidney Stone Consortium is funded through collaboration between NCATS and National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). The study sponsor had no role in study design; collection, analysis, and interpretation of data; writing the report; or the decision to submit the report for publication.

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**Take home message (synopsis):**

High urinary 2,8-dihydroxyadenine (DHA) excretion was observed in all patients with APRT deficiency while DHA was not detected in urine samples from *APRT* heterozygotes and healthy individuals, suggesting that our robust and reliable UPLC-MS/MS urinary DHA assay can be added to the list of diagnostic tests for APRT deficiency.

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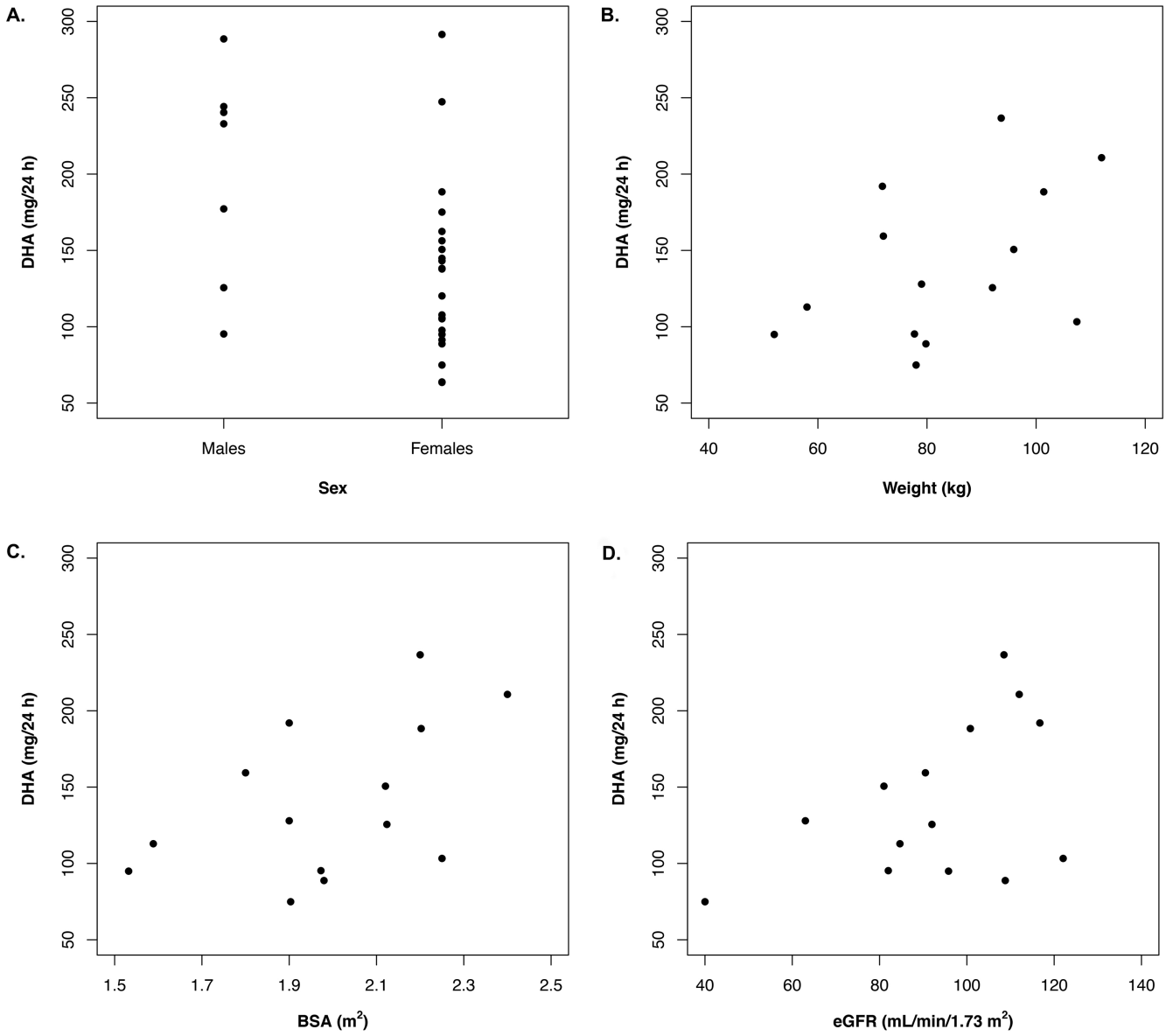



**Figure 1.**

Scatterplot of 24-hr urinary DHA with (A) sex (p = .03), (B) weight ($r_s$ = 0.36, p = .203), (C) BSA ($r_s$ = 0.36, p = .208), and (D) eGFR ($r_s$ = 0.35, p = .227) in patients with adenine phosphoribosyltransferase deficiency who were not receiving xanthine oxidoreductase inhibitor treatment.

Abbreviations: BSA, body surface area; DHA, 2,8-dihydroxyadenine; eGFR, estimated glomerular filtration rate.

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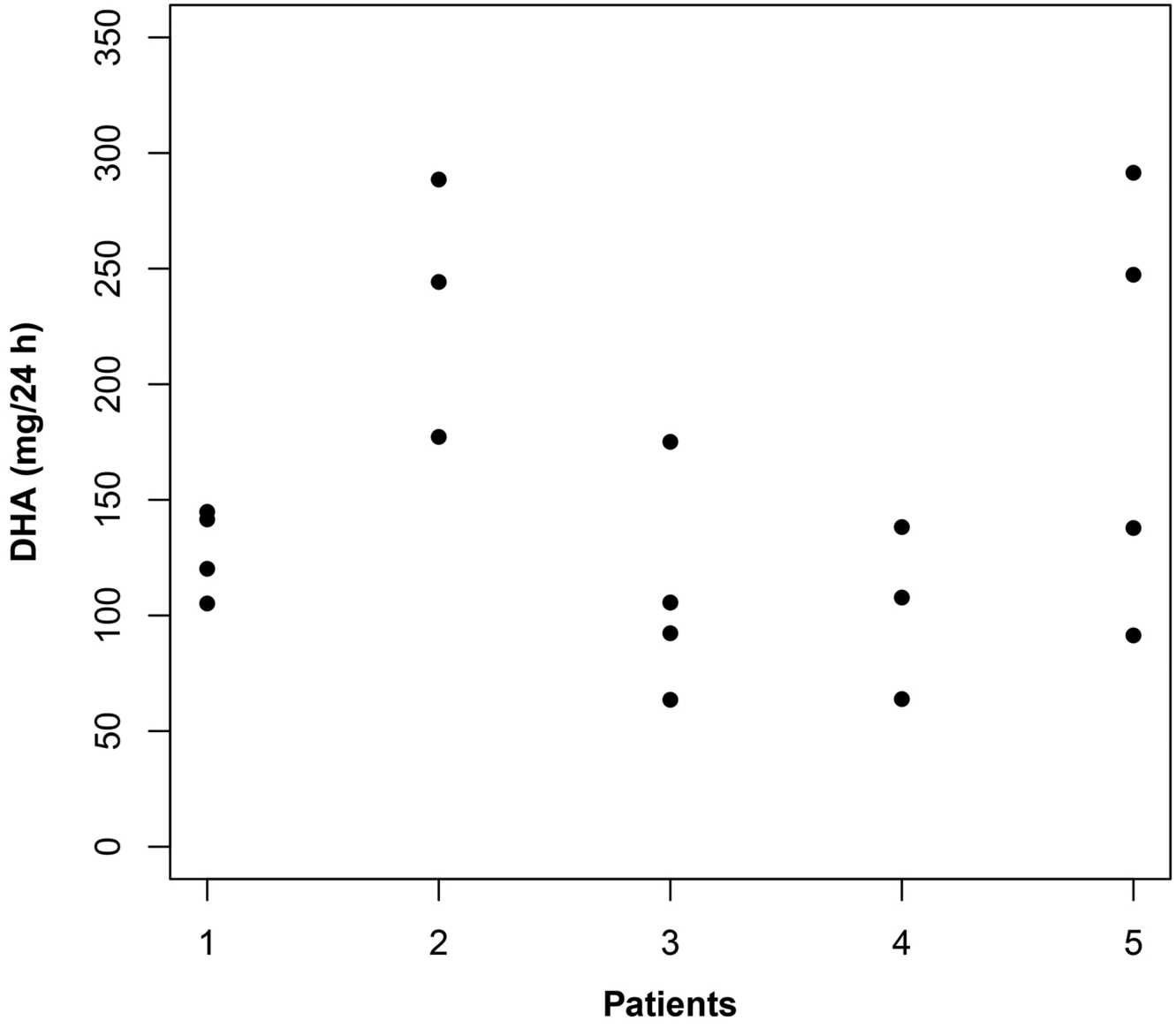



**Figure 2.**

Twenty-four hour urinary DHA excretion in five patients with adenine phosphoribosyltransferase deficiency who were not on xanthine oxidoreductase inhibitor treatment. For each patient, measurements were carried out in urine samples obtained at different time points.

Abbreviations: DHA, 2,8-dihydroxyadenine.

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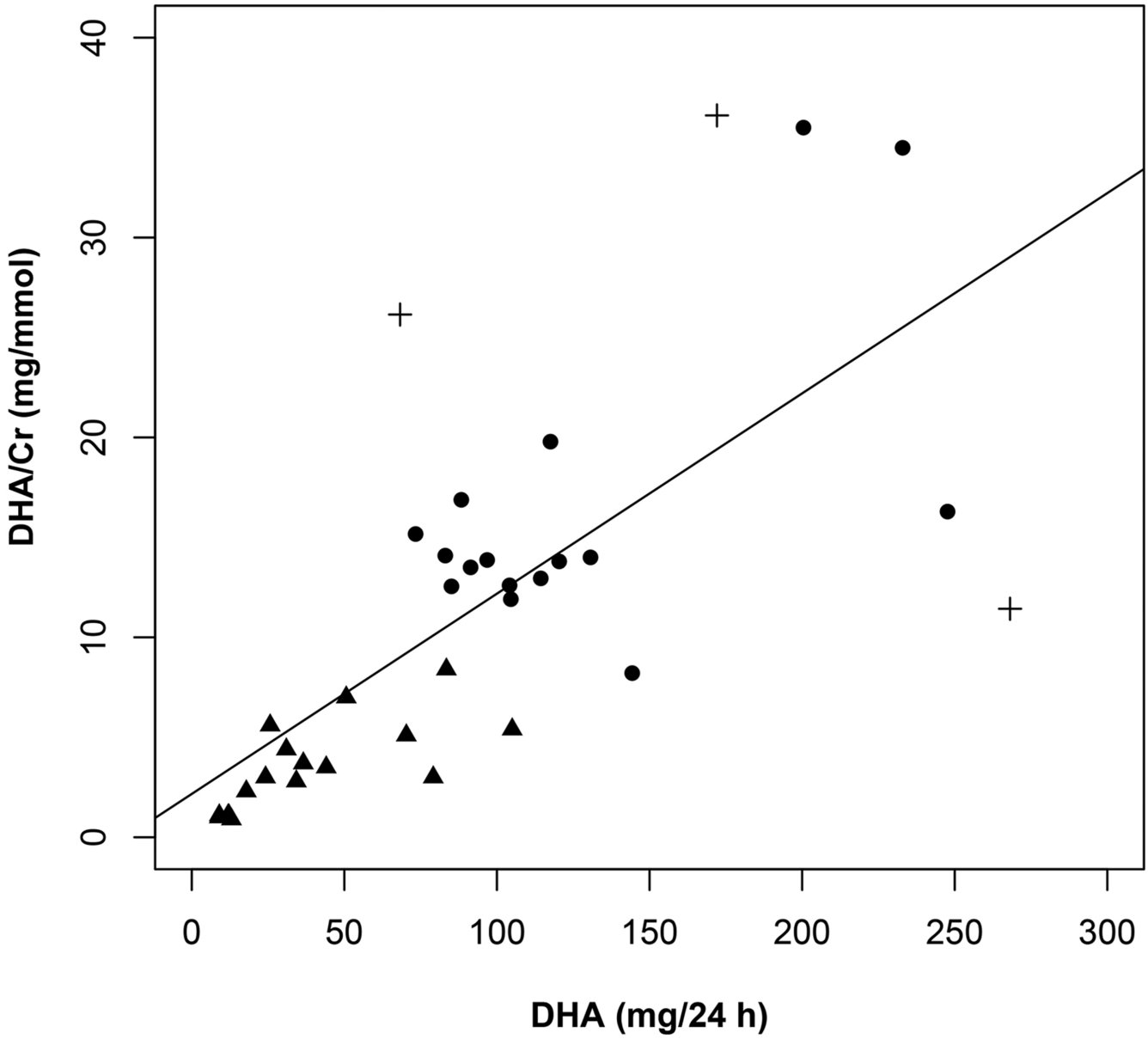



**Figure 3.**

Scatterplot of 24-hr urinary DHA excretion versus DHA/Cr ratio in first-morning void urine samples from patients with adenine phosphoribosyltransferase deficiency who were treated (▲) or not treated (•) with xanthine oxidoreductase inhibitor ($r_s$ = 0.84, p <0.001).

*Outliers (+) were excluded from the correlation analysis.

Abbreviations: Cr, creatinine; DHA, 2,8-dihydroxyadenine.

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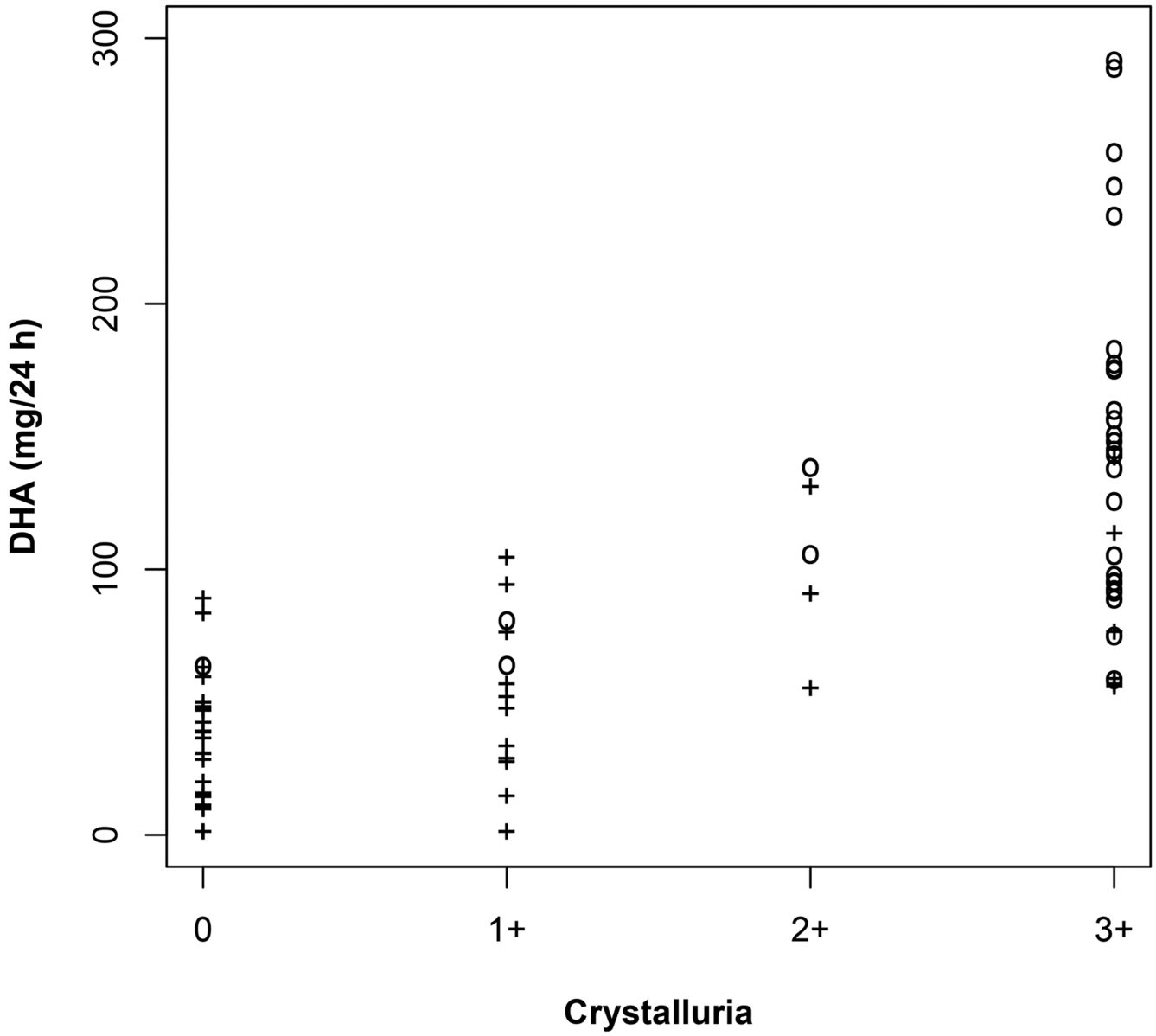



**Figure 4.**

Twenty-four hour urinary DHA excretion versus microscopic DHA crystalluria in patients with adenine phosphoribosyltransferase deficiency who were treated (+) or not treated (○) with xanthine oxidoreductase inhibitor ($r_s$=0.810, $p < 0.001$). No patient had 4+ crystalluria. Abbreviations: DHA, 2,8-dihydroxyadenine.

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**Table 1.**

Urine samples from patients with adenine phosphoribosyltransferase deficiency available for analysis

| | Homozygotes | | | |
|---|---|---|---|---|
| | Paired | | Not paired | |
| **Not on XOR inhibitor therapy** | **Samples (n)** | **Patients (n)** | **Samples (n)** | **Patients (n)** |
| **24-hour samples** | 19 | 11 | 12 | 7 |
| **First-morning void samples** | 19 | 11 | 25 | 13 |
| **Random samples** | | | 24 | 4 |
| **On XOR inhibitor therapy** | | | | |
| **24-hour samples** | 17 | 8 | 43 | 21 |
| **First-morning void samples** | 17 | 8 | 121 | 33 |

Abbreviations: XOR, xanthine oxidoreductase.

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**Table 2.**

Urinary DHA excretion in patients with adenine phosphoribosyltransferase deficiency (not on XOR inhibitor therapy), heterozygotes and healthy control subjects

| | Homozygotes | Heterozygotes | Healthy control subjects |
|---|---|---|---|
| ***Number of participants*** | 19 | 4 | 10 |
| **Age, years** | 35.4 (16.1-67.0) | 47.4 (36.6-56.6) | 25.0 (23.5-30.1) |
| **eGFR, mL/min/1.73 m$^2$** | 101 (14-131) | 94 (55-100) | 106 (78-125) |
| **Weight (kg)** | 79.4 (52-112) | 77.2 (62-116) | 72.4 (58-93) |
| **BSA (m$^2$)** | 1.9 (1.5-2.4) | 1.9 (1.7-2.5) | 1.8 (1.7-2.2) |
| ***24-hour urine collections*** | | | |
| **Number of participants** | 14 | 4 | 10 |
| **Number of samples** | 31 | 4 | 10 |
| **DHA/24 hr, mg** | 138.2 (63.5–291.5) | BLQ* | BLQ* |
| **Urine creatinine, mmol/kg/24 hr** | 0.15 (0.10-0.20) | 0.18 (0.15-0.24) | 0.15 (0.10-0.21) |
| ***First-morning void urine samples*** | | | |
| **Number of participants** | 19 | 4 | 10 |
| **Number of samples** | 44 | 4 | 10 |
| **DHA/Cr ratio, mg/mmol** | 12.7 (3.8–37.2) | BLQ* | BLQ* |
| ***Random urine samples*** | | | |
| **Number of participants** | 4 | NA | NA |
| **Number of samples** | 24 | | |
| **DHA/Cr ratio, mg/mmol** | 15.5 (10.4-19.3) | | |

Abbreviations: BLQ, below limit of quantification (<100 ng/mL); BSA, body surface area; Cr, creatinine; DHA, 2,8-dihydroxyadenine; eGFR, estimated glomerular filtration rate; NA, not available; XOR, xanthine oxidoreductase;

* $p<0.005$